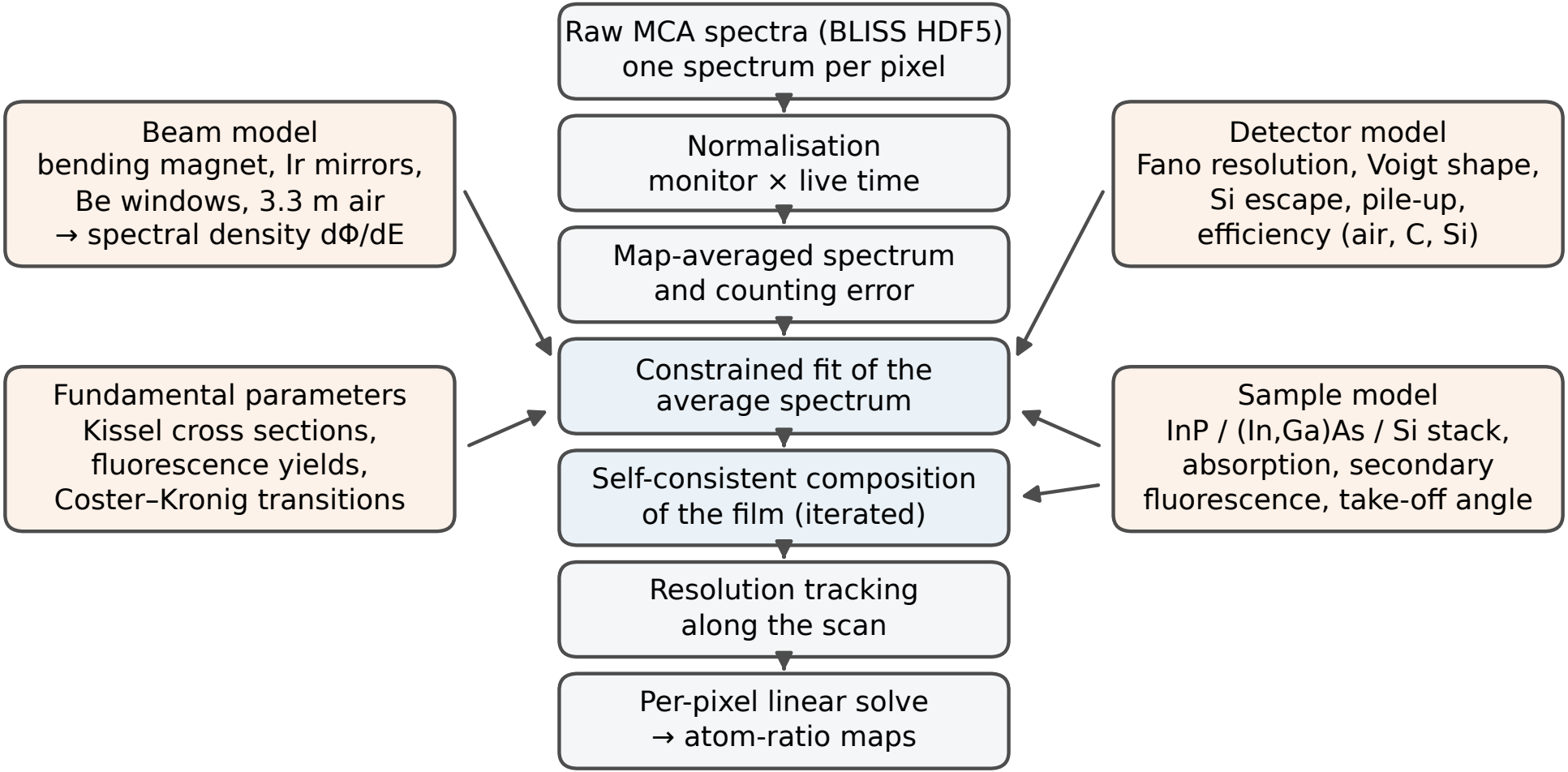
Raw MCA spectra (BLISS HDF5)
one spectrum per pixel
Normalisation
monitor × live time
Map-averaged spectrum
and counting error
Constrained fit of the
average spectrum
Self-consistent composition
of the film (iterated)
Resolution tracking
along the scan
Per-pixel linear solve
→ atom-ratio maps
Beam model
bending magnet, Ir mirrors,
Be windows, 3.3 m air
→ spectral density dΦ/dE
Fundamental parameters
Kissel cross sections,
fluorescence yields,
Coster–Kronig transitions
Detector model
Fano resolution, Voigt shape,
Si escape, pile-up,
efficiency (air, C, Si)
Sample model
InP / (In,Ga)As / Si stack,
absorption, secondary
fluorescence, take-off angle